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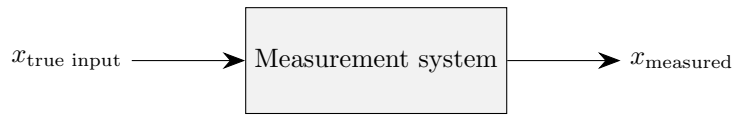
1 Uncertainty Analysis

When we report an experimental result, we should report:

$$\begin{aligned}x &= \bar{x} \pm U_x \\ \text{or} \\ \mu_x &= \bar{x} \pm U_{\bar{x}}\end{aligned}$$

where, In 95% of x $\begin{cases} U_x = \text{expanded uncertainty of } x \\ U_{\bar{x}} = \text{expanded uncertainty of } M_x \end{cases}$
This tolerance term is U_x

For measurands that vary continuously, one cannot measure their values exactly, i.e, there will always be uncertainty



$$x_{\text{measured}} = \langle x \rangle + \alpha_x + \text{Random Errors} + \text{Systematic Errors}$$

Where,

- $\langle x \rangle = \mu_x = \text{population mean}$
- $\alpha_x = \text{fluctuation}$
- Random Error - Intrinsic to the measurement system itself (ex. noise)
 - Generally different for each successive measurement
- Systematic error (symbol β) - Occurs in the same way each time a measurement is made (ex. A thermometer always reads 1 $^{\circ}\text{C}$ too high)
- Elemental error
 - An identifiable error source
 - Classifies as either random or Systematic Also arise from environment variation

Examples:

accuracy	systematic	β
hysteresis	systematic	β
installation	systematic	β
linearity	systematic	β
repeatability	systematic	β
noise	random	ϵ
environmental	random	ϵ
resolution	random	ϵ
digitization	systematic	β
reading	random	ϵ
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